PROGRESS REPORT

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Section: CSIS4495 - 002

Project's name: 30-Day Readmission

Date	Number of Hours	Description of work done
Sep 17, 2026	1.5	I reviewed proposal draft; evaluated Streamlit/Power BI/Tableau stack against a single web application; decided on React + FastAPI + PostgreSQL
Sep 19, 2026	~1	Produced UI mockups of five application screens: cohort intake, discharge queue, patient detail with SHAP explanation, model and fairness report, threshold and cost-benefit
Sep 19, 2026	~2	Rewrote proposal for clarity; updated the timeline for the revised stack
Sep 19, 2026	30mins	Specified runtime inputs (replayed test cohort, CSV upload, single-patient form); documented the prediction-timing constraint
Sep 22, 2026	1	Methodology review; added patient-level grouped splitting,

		probability calibration, race in the subgroup analysis, cohort and ethics caveats, and a contingency plan
Sep 22, 2026	~1	Condensed proposal to 5 pages against the 6-page limit
Sep 23,2026	1.5	Created Github…
Oct, 4 2026	~2	Reviewing Dataset, and Reviewing the machine learning course
Oct 6 2026	~2	Identify imbalance target
Oct 6	~4	Do 3step cleaning data, do pipeline, dropping null values, delete some rows, grouping rows…
Oct 7	~ 5,7	I stratified split function like train_test_split data, set up PostgreSQL in Docker with a star-schema database and SQL views for analysis
Oct 9	~6,7	Exploratory analysis data
Oct 11	~3.4	Exploratory analysis data

Summary Description of work done during this reporting period: I mostly forgot how to do the machine learning, I tried to figure it out, Then one thing I wanna tell you that in my previous report, I just say about doing SQL but after searching out carefully, I use PostgreSQL because of this I can get data and collect data from user to do the website easier, and I also bought claudeAI plus to help me some stuff like easy to understand and better.

Over the first four weeks, I research the dataset, I research about the machine language I habe to use and how can I combine these things to be a website as you told me, and mostly from last week, I started to do the project, I set up the project, completed the proposal, and built the data foundation for predicting 30-day hospital readmission in diabetic patients, using the UCI Diabetes 130-US Hospitals dataset.

I also separate weeks into the small steps so just easier for you and me to check and see what have been done:

- **Setup and proposal (Weeks 1–2):** created the GitHub repository and submitted the project proposal. This part was just for the discussion step in the class between us.
- **Data understanding (Week 3):** profiled 101,766 hospital encounters. Identified an imbalanced target (11.2% readmitted within 30 days), heavy missing data in some columns, but for some code I forgot from what I learnt in machine learning so I reviewed the course and used claude to do some parts, and repeat patients who could cause data leakage.
- **Data pipeline (Week 3):** before moving forward to this step, I cleaned up the data by dropping null values, delete some rows that have the same data, do cleaning steps then I built a reusable cleaning and feature-engineering module shared by training and the API, giving a clean dataset of 94,660 encounters with an 11.4% readmission rate. Split it 60/20/20 into train, validation and test sets, grouped by patient, and verified that no patient appears in more than one set.
- **Infrastructure (Weeks 3–4):** set up PostgreSQL in Docker with a star-schema database and SQL views for analysis. Scaffolded a FastAPI backend and a React frontend, and added unit tests (all passing).
- **Exploratory analysis (Week 4):** produced 7 charts of readmission rates by key factors, and ranked 27 features by univariate AUC but I didn't have time enough to write the description, I will do it later. Researching and showing the strongest predictors are prior inpatient stays (AUC 0.613; readmission rises from 8.4% with no prior stays to 32.0% with four or more), total prior visits and discharge disposition. Demographics (race, gender) showed almost no individual predictive signal.

Here we are in the current status: the data pipeline, database and EDA charts are complete. Next is writing up the EDA.

Repo Check in of Implementation completed:

This are these files Im working with recently, including all steps that I listed above

- DocumentsAndReports/ThienN_Proposal.pdf
- DocumentsAndReports/ThienN_ProgressReport1.pdf
- Implementation/docker-compose.yml (PostgreSQL in Docker)
- Implementation/notebooks/ (data profiling and EDA notebooks)
- Implementation/ml/ (shared cleaning and feature-engineering module; patient-grouped 60/20/20 split; database loading)
- Implementation/sql/ (star schema and SQL views for analysis)

- Implementation/api/ (FastAPI backend scaffold)
- Implementation/web/ (React frontend scaffold)
- Implementation/tests/ (unit tests, all passing)
- Implementation/results/ (7 EDA charts; feature ranking by univariate AUC)

AI Use Section:

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AI Tool Name	**Version, Account Type**	**Specific feature for which the AI tool was used**	**Value Addition** ***What value did you add over and above what AI did for you? In your own words***
Claude	plus		I used to search about PostgreSQL because I have never done with it before professor
claude	plus		There is a step when I do python it the file 02_eda.ipynb, but it shows errors a lot and I used claude to make it work, my code wasn't wrong, it was from different computers and localhost, so It gives me this line : import os; os.chdir("..") . and Its working
Claude	plus		I used it to figure out cuz there are errors in the 01_inspect.ipynb file, and I figured out by creating another file named Data, and I put the dataset in there, so I file can read it and run it successfully
Claude	plus		In the features file, I also used claude to know how can I group the dataset together to, I used how to get the commands like examples: ADMISSION_SOURCE_GROUP={}…
Clause	plus		I also used claude to show me how can I create the postgreSQL and it gives me some virtual environments and steps to set it up professor

▪ ***Appendix:*** *Prompt history must all be saved in the appendix*

The Grading Rubric for the Progress Report

Criteria	**Grading**

Work Logs • The consistency of the work logs • The details and description of the work logs • The granularity of the work logs	2 points
Summary Description of Work • The complexity and technical merit of the work done • The documentation of work done and issues logged in the repo	2 points
AI Use Section • AI Use section is specific and detailed • Prompt is included in the appendix	2 points
Repo Check In • Frequency of work completed • Frequency of repo check in • Quality of work completed	4 points